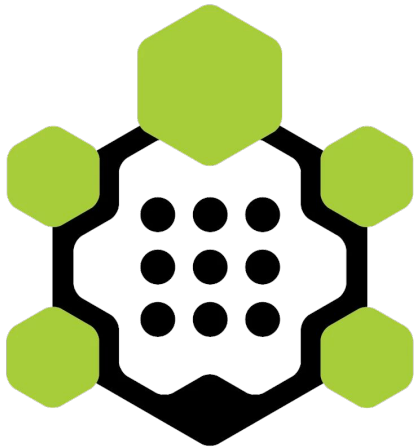
Robot-Guided Home Rehabilitation

Monitoring and therapist feedback between sessions

Team 105

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Problem

- Gait rehabilitation depends on walking exercises done at home, alone.
- Patients lose rhythm, drift into asymmetry as they tire and cut sessions short.
- The physiotherapist never sees any of it — only what the patient remembers at the next appointment.

Solution

A robot that supervises home session between appointments.

- **Guides** — sets the route and the rhythm of the walk
- **Measures** — cadence, speed and symmetry, from leg movement at shin height
- **Cues** — speaks a correction in the moment
- **Reports** — turns each session into data the physiotherapist can act on

Proof of Concept

A TurtleBot3 that:

- sets the walking rhythm with an audible beat
- measures cadence, speed and symmetry from leg movement
- speaks a correction when the patient drifts off the rhythm or stops
- publishes a session summary to a dashboard

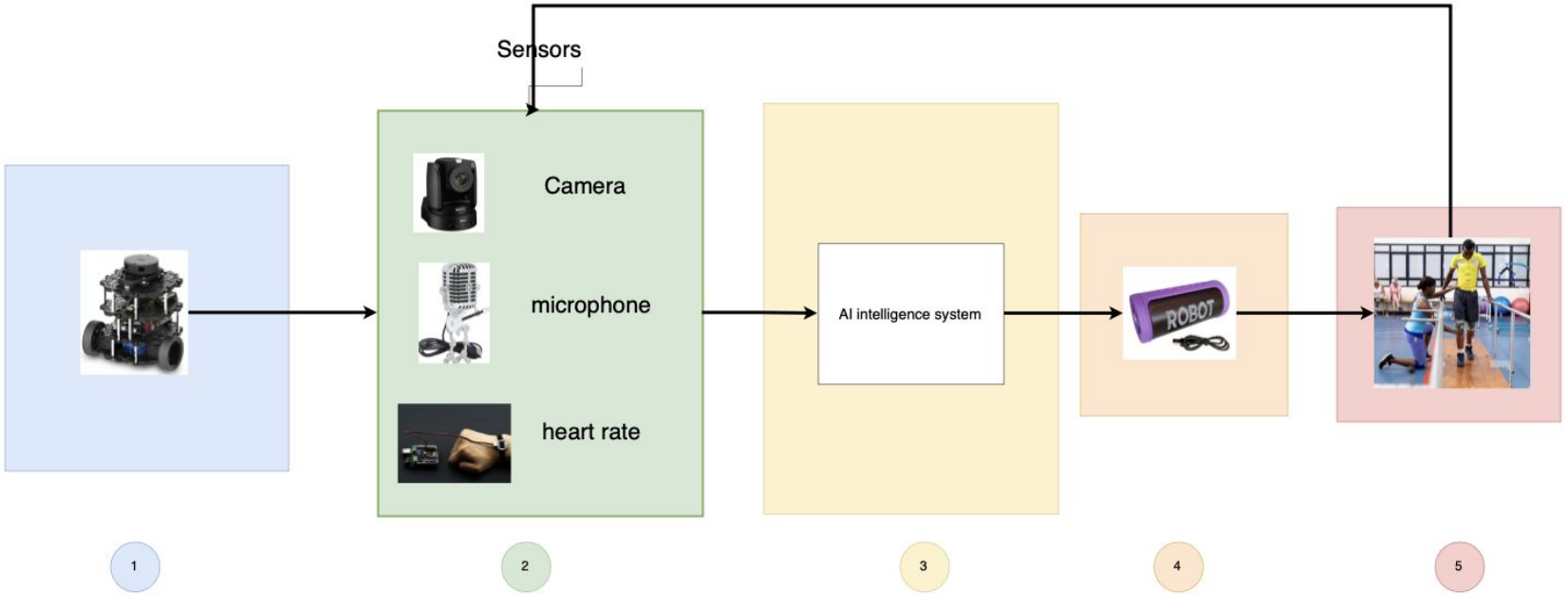
Scope

- Healthy volunteer, flat corridor, one exercise, capture zone is ~7 m.
- Not clinical. No physical support.
- Between physiotherapy sessions, not instead.

How we'll evaluate

1. Detection latency — how quickly the robot notices a stop
2. Measurement accuracy — cadence against a manual step count
3. Cue reliability — does the right cue fire, across repeated walks

Architectural Design



Thank You!